

01_Data Relationships & Schema Understanding

Schema Overview

The Olist dataset follows a relational database structure where transactional, descriptive, and supporting tables are connected using common identifiers such as `customer_id`, `order_id`, `product_id`, and `seller_id`. These relationships help track the complete business flow from customer purchases to product delivery and customer feedback.

Core Business Flow

Customers

- Place Orders
- Orders Contain Products
- Products Are Sold By Sellers
- Payments Are Made For Orders
- Orders Receive Reviews
- Deliveries Occur Across Geographic Locations

Table Relationships

Parent Table	Child Table	Relationship
customers	orders	One customer can place many orders
orders	order_items	One order can contain many products/items
products	order_items	One product can appear in many orders
sellers	order_items	One seller can fulfill many order items
orders	payments	One order can have multiple payment records
orders	reviews	One order can receive a review
products	category_translation	Product categories can be translated

Fact Tables vs Dimension Tables

Type	Tables
Fact Tables	orders, order_items, payments, reviews
Dimension Tables	customers, products, sellers
Supporting Tables	geolocation, category_translation

Initial Analytical Understanding

The relational structure of the dataset supports customer, product, revenue, operational, and customer experience analysis. Combining transactional tables with descriptive tables enables deeper business intelligence and analytical exploration across customer, revenue, operational, and product domains.